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Mathematisches Seminar
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Practical Exercise to Numerical Optimization

We consider the following unrestricted optimization problem:

$$\text{(UOP)} \qquad f(x) \;:=\; (x_1^2 - x_2)(x_1^2 - 3x_2) \;\to\; \min!, \qquad x \in \mathbb{R}^2.$$

Exercise 1:

The corresponding one-dimensional restriction with respect to a direction $d \in \mathbb{R}^2$, $\|d\|_2 = 1$, is given by

$$\varphi_d(t) \;:=\; f(x + td).$$

Write code (or download) that determines local minima of φ_d with $x = d$. One possibility is to solve the first-order necessary condition

$$\varphi_d'(t) \;=\; 0$$

by a descent method. Derive φ_d' and φ_d'' (by hand). You should obtain the following result: The point $t = -1$ is a local minimum of φ_d for any d, i.e. $x + td = (0, 0)$.

Restricted Problem

The restricted optimization problem we consider is given by

$$f(x^*) = \min\{f(x) : x \in X\} \quad \text{(ROP)}$$

with $X := \{x \in \mathbb{R}^2 : g(x) \leq 0\}$ and

$$g(x) := -x_1 + x_2 + c.$$

Exercise 2:

Start writing the code without any constraints (UOP). Use a simple gradient search $d := -\nabla f(x)$ and the following types of step sizes:

- constant step sizes, and
- Armijo rule.

It will be useful to perform the first tests with a function having a global minimum, e.g.

$$f(x) = \frac{1}{10}x_1^2 + x_2^2.$$

Exercise 3:

Use the penalty method (with penalty parameter $\alpha > 0$) in order to consider inequality constraints. Apply your algorithm to get solutions in dependence of $\alpha = 2^k$, $k = 0, \ldots, 4$. You may use the solution for $\alpha = 2^k$ as initial guess for the iteration for $\alpha = 2^{k+1}$. Use a simple gradient search $d := -\nabla f(x)$ and the step sizes from the previous exercise. Do you obtain admissible solutions?